

Pattern Arithmetic — Quick Reference

Extracted from Pattern_Arithmetic.v15.15. Section numbers refer to that document.

Changes since the v15.11 edition of this reference: the introduction now opens with a stated Thesis (§0 below); T 's dimensional generalization is joined by an equivalent one for Offset's transport, both verified rather than merely claimed, up to $N_{\text{dim}} = 50$; and the Enclosure gained a fourth named convention, the allowance ϵ_{allow} , for treating nearby checkpoints as the same freeze.

0. The Thesis (context, not a formal rule)

The introduction now states the paper's governing idea directly, before any axiom:

Mathematics need not be performed on numbers alone. Wherever a domain supplies a unit of truth and a small set of rules by which such units combine, obeyed consistently, the unit and its rules together form an arithmetic of that domain.

Color, musical notes, Chladni figures, valence chemistry, and the mitochondrion (joined without dissolving — a natural-world instance of Assembly) are offered as *evidence* that such arithmetics exist elsewhere, not as domains this paper models. Ice's six-fold growth is named as one such domain arithmetic in its own right — and, by Axiom 0, exactly the kind refused a map onto this paper's sphere. This paper states its own rules as axioms rather than discovering them; the “settle toward stability” half of the thesis appears only in miniature, in the Enclosure (§9).

1. The Axioms

#	Name	One-line summary
0	Admissibility of Sorts	Before any operator can be used, a sort τ must earn the right to a geometric reading — the table of who-may-use-what operators comes <i>after</i> this, not instead of it.
1	The Normalized Unit Truth State ($r \leq 1$)	A pattern is one typed point, not a field: $x = (r, \hat{n}, \chi, \tau)$, $r \in [0, 1]$, $\hat{n} \in S^2$, $\chi \in [0, 2\pi)$. A bowl is a separate finite set of such atoms.
2	The Absolute Null State ($r = 0$, Black)	Vanishing strength means direction and phase are unused — an empty seat, not a field that happens to be zero everywhere.

#	Name	One-line summary
3	Saturation (White)	The upper pole of the same bounded stage, not an infinite norm. Reached either as a single full-spectrum identity or as a bowl with many directions filled — neither pushes any single arrow past $r = 1$.
4	Gather and Remove ($\oplus, \ominus$)	$\oplus$ is set union on bowls — it collects, it does not blend: $A \oplus A = A$. $\ominus$ removes one member. Set union needs no dimension, so this is unaffected by anything below.
5	Mix as a Reading ($\boxplus$)	A phasor operator: combines χ among states that already share a direction $\hat{n}$. Reads a bowl into one blended state; never moves an arrow onto a new axis. Lossy and non-associative under the cap — see §3 below. Dimension-blind wherever it applies, since it never leaves the shared axis.
(<i>unnumbered</i>)	Rotation preserves the unit	$\mathbf{R} \in \text{SO}(3)$ turns $\hat{n}$, leaves r, χ, τ untouched. The geometric engine T is built from, not a second kind of blend.
6	Assembly ($\bowtie$)	The fourth verb: given two intact units, how strongly do they belong together and where do they touch? $A \bowtie B = (A, B, \kappa_{AB}, U_{AB})$ — both units survive unchanged; nothing is fused. Never repels in any dimension: the alignment bracket $3 + \hat{n}_A \cdot \hat{n}_B$ stays in $[2, 4]$ regardless of N_{dim} .
7	Informational Vectors & Derived Per-Atom Metrics	An atom may be a <i>bundle</i> of N informational vectors rather than one — still teaching atoms under one closure, not a new species of unit. Defines V_{derived} and the seven scalar metrics of §3.

Two further operators are built on these axioms but are not themselves axioms — both now generalize beyond three dimensions, verified rather than merely claimed:

- **Transform** (T) — §3.4: an actor w rotates an occupant x about w 's own axis by $\alpha(r_w) = \pi r_w$. Beyond $N_{\text{dim}} = 3$, the actor's bundle supplies $N_{\text{dim}} - 2$ mutually orthogonal arrows to fix the axis (see the ladder in §3 below); checked invertible and non-commutative up to $N_{\text{dim}} = 50$, exact at $N_{\text{dim}} = 3$.
- **Offset** — §4.1: plants a new seat from three seats via log/exp/transport. The transport is now written for general dimension too: a rotation within the plane the two seats themselves span, needing no bundle, exact at $N_{\text{dim}} = 3$ against the original cross-product form.

2. Original Metrics — the base unit (Axiom 1)

The four coordinates of a single teaching atom. Nothing here is derived from anything else; every other quantity in this paper is built from these.

Symbol	Domain	What it answers
r	$[0, 1]$	How strong / how confidently held is this unit?
$\hat{n}$	S^2 (equivalently (θ, ϕ))	What direction — the semantic address?
χ	$[0, 2\pi)$	What internal phase/mood/hue?
τ	$\{\text{word, num, color, ...}\}$	What sort — which operators may legally act on it?

$N = 1$ is the default: one atom, one arrow. Axiom 7 is what allows $N > 1$.

3. Derived Metrics — built from the base unit (Axiom 7)

All of these presuppose a **bundle**: N informational vectors $v_i = r_i \hat{n}_i$, each an instance of the Axiom 1 unit, held under one closure.

Symbol	Formula	What it measures	Requires
V_{derived}	$\sum_{i=1}^N v_i$	The raw, uncapped vector sum of the bundle. Reads as an effective $(r, \hat{n})$ pair when a single vector is needed — but is <i>not itself a unit</i> and needs a cast before acting as T 's actor w .	any $N \geq 1$
S_{atom}	$\sum_i r_i$	Atom strength. The correct denominator for atom-level coherence, $ V_{\text{derived}} /S_{\text{atom}}$.	any N
$\mathcal{V}_{\text{atom}}$	$\sum_i r_i^3$	Atom volume. Strictly $< S_{\text{atom}}$ whenever any $r_i < 1$ — a forced consequence of the cap, not an assumption.	any N
γ_{atom}	$\mathcal{V}_{\text{atom}}/N$	Mean vector volume. Average per-vector share of the total.	$N \geq 1$
$\mathcal{E}_{\text{atom}}$	$\sum_i r_i^2$	Information energy. Stands on its own (not squared strength) — the bundle's intrinsic energy <i>before</i> asking how its vectors align.	any N
α_{atom}	$\frac{\sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)}{\sum_{i < j} r_i r_j}$	Internal alignment. Strength-weighted mean cosine similarity across all pairs — separates active cancellation (< 0) from mere non-reinforcement, which coherence alone cannot. Checked at $N_{\text{dim}} = 5, 20$ as a dot-product identity: dimension does not enter it.	$N \geq 2$

Symbol	Formula	What it measures	Requires
σ_{atom}	$\frac{N^2}{N-1} \text{Var}(s_1, \dots, s_N),$ $s_i = r_i^3 / \mathcal{V}_{\text{atom}}$	Volume concentration. 0 when spread evenly, 1 when one vector holds it all, at every N .	$N \geq 2$ ($= 0$ at $N = 1$)
$\sigma_{\text{shell}}(w)$	$\frac{K^2}{K-1} \text{Var}(p_1, \dots, p_K),$ bins of width w (default 0.1)	Shell concentration. Catches a blind spot σ_{atom} has by construction: many vectors independently landing at the same magnitude.	≥ 2 occ. shells ($= 1$ at $K = 1$)

The dimension ladder (§3.4) — how many arrows a bundle needs before it may act as T ’s actor w beyond three dimensions:

Space	Sphere	Fixed part (the “axis”)	What turns
$\mathbb{R}^2$	S^1 , a circle	a point, dimension 0	one plane
$\mathbb{R}^3$	S^2	a line, dimension 1	one plane
$\mathbb{R}^4$	S^3	a plane, dimension 2	one plane
$\mathbb{R}^5$	S^4	a 3-D space, dimension 3	one plane
$\mathbb{R}^{N_{\text{dim}}}$	$S^{N_{\text{dim}}-1}$	dimension $N_{\text{dim}} - 2$	one plane

A teaching atom ($N = 1$) is therefore a legal actor only at $N_{\text{dim}} = 3$; beyond that, the bundle must supply $N_{\text{dim}} - 2$ **mutually orthogonal** arrows. Verified invertible and non-commutative up to $N_{\text{dim}} = 50$.

4. Other named derived quantities (by operator)

For completeness — these are built *within* a specific axiom/operator rather than being general per-atom metrics.

Symbol	Formula	Lives in	What it is
U_{AB}	$-\frac{r_A r_B}{2} \cdot \frac{3 + \hat{n}_A \cdot \hat{n}_B}{(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}},$ $\varepsilon_{AB} = r_A + r_B$	Axiom 6	Assembly’s interaction energy. Always < 0 for nonzero strengths — never repels, in any dimension.
κ_{AB}	$\hat{n}_A + \hat{n}_B$ (unnormalized)	Axiom 6	Assembly’s contact point; defined at every angle, including antipodal ($= \mathbf{0}$).
$\alpha(r_w)$	πr_w	§3 (T)	Maps an actor’s strength to a rotation angle; full strength ($r_w = 1$) gives a 180° turn.

Symbol	Formula	Lives in	What it is
$z_B, \boxplus(B)$	$\sum_{x \in B} r_x e^{i\chi_x}; \text{result} = (\min(1, z_B), \hat{n}, \arg z_B, \tau)$	Axiom 5	Mix's one-step bowl reading — the reason binary Mix is non-associative under the cap.
$v, \text{transport}, \hat{n}_*$	$v = \log_{\hat{n}_m}(\hat{n}_k);$ $\hat{n}_* = \exp_{\hat{n}_w}(\text{transport}_{m \rightarrow w}(v))$	§4.1	The copied walk: log the source pair, transport the tangent vector, exponentiate at the new seat. At $N_{\text{dim}} = 3$, transport is the original cross-product rotation about $\hat{n}_m \times \hat{n}_w$; in general, it is a rotation within the plane $\hat{n}_m, \hat{n}_w$ themselves span — the two seats name the turning plane, so no bundle is needed. The two forms agree exactly at $N_{\text{dim}} = 3$.
F_i	$-\nabla_{\vec{P}_i} \sum_{j \neq i} U_{ij}$	§9	Force on a seated unit — descended without mass or momentum.
freeze condition	$\max_i F_i < \eta$	§9.2	A configuration below threshold η is a <i>checkpoint</i> : it may be frozen and taken as the accepted solution — not exact rest, since the pull never reaches zero at finite distance, so left longer every configuration drifts toward one pile.
$\vec{P}_{c,k}$	$\frac{\sum_{i \in G_k} r_i \vec{P}_i}{\sum_{i \in G_k} r_i}$	§9.3	A camp's body — the strength-weighted centre of its members, read <i>after</i> motion, not sought during it.
ϵ_{allow}	(named, not derived)	§9.3	Allowance. Two checkpoints whose bodies differ by less than ϵ_{allow} are read as the same freeze; a fourth named convention alongside η, δ , and the census cut — not a truth of U_{AB} . A camp read this way is still only a checkpoint on its members, never a new unit in its own right.

This reference is a summary aid only; the equation numbers and full derivations live in Pattern_Arithmetic_v15_15.tex.